About Tekion:

Tekion is a cloud-built platform focussed on transforming consumer experience in the automotive industry. The company uses advanced machine learning and AI capabilities for a personalised and seamless consumer experience.

Online Round:

Online round consists of 2 sections

MCQ: 20 questions (DBMS, CN, Dsa, OOPS)

DSA:

Problem 1:

You are given two arrays of equal length: process[] and state[].The process [] array contains alphanumeric process identifiers such as "A1", "B102", etc., while the state [] array contains only "Login" or "Logout", indicating the corresponding action for each process at the same index. A process's activation counts increases by 1 whenever it logs in, and decreases by 1 whenever it logs out, but the count can never go below 0. After processing all the events in order, you are given an integer m, and your task is to determine which processes remain active in the system with an activation count strictly less than m. process = ["B1", "A1", "B1", "B1", "A1"] state = ["Login", "Login", "Login", "Login", "Logout"] m = 2

Problem 2:

You are given a matrix of size n × m. You must pick exactly (m-1) rows. Then construct a "set" (really, an array of column maximums): For each column j, take the maximum value among the chosen rows in that column.
The score = the minimum value in that array of column maximums.
Goal: Find the maximum possible score over all choices of (m-1) rows.

I did the MCQ well. I was able to solve the first question completely and in second question I passed 8/15 with a greedy approach. The results were announced on the same night

Interview:

Round 1 (Virtual) - Technical

The interviewer began by asking me to introduce myself. After that, I was asked to open a compiler of my choice and solve the following DSA problems.

Problem 1:
Given integers n and k, return an array containing all possible numbers with n digits such that the absolute difference between every pair of adjacent digits is k. The number should not contain any leading zeros.

My Approach:

- I first explained a brute-force solution where I would generate all possible n-digit numbers and check whether they satisfy the condition.
- I then provided an optimal solution using recursion, where only valid numbers were generated without exploring invalid possibilities.
- The interviewer tested my recursive solution with a few test cases, and it worked correctly.

After this, he asked me about my favorite data structure. I mentioned that it is the **stack**, and explained the reasons behind my preference. I was expecting a stack-related problem next, but the interviewer moved on to a different question.

Problem 2: Subarray Sum Equals K

My approach:

- I first explained a brute-force solution by generating all subarrays and checking whether their sum equals k. I also discussed the time and space complexity of this approach.
- Then, I provided an optimal solution using a hash map to keep track of prefix sums.
- The interviewer tested my solution with a case where the array contained negative numbers. My solution also worked correctly for this scenario, and the test case passed.

At the end of the interview, I asked the interviewer some questions about the work environment and also requested tips for the upcoming rounds.

Round 2: (Offline) - Technical

- The interviewer introduced himself, spoke about his experience at Tekion, and then asked me to introduce myself.
- In my introduction, I mentioned that I had participated in competitive coding competitions. This caught his attention, and he asked me more about it.
- He asked what I knew about Tekion.
- He also asked whether Tekion was in my wishlist of companies. I replied yes and explained why.
- Then, he asked me a series of technical questions:
  - What are microservices?
  - What are asynchronous and synchronous functions?
  - What is concurrency, and how it is different from parallelism?
  - What is a garbage collector, and why is it useful?
  - What scheduling algorithms do I know, and how can they be used to achieve concurrency?

- **DSA Problem:** [Missing Number]
  I explained three different approaches to solve this problem and discussed the time and space complexity for each approach.
- He then moved to computer networks and asked:
  - What is computer networks?
  - What protocols do you know?
  - Explain the SMTP protocol.
  - What are TCP and UDP?
  - What is the difference between TCP and UDP, and where are they useful?
  - Which protocol would you use if you had to announce an upcoming disaster to a group of people?
- At the end, he asked whether I had any questions for him. I said yes, and I asked him to tell me more about Tekion and also about the role of a software engineer at Tekion.

Round 3: (Offline) – hr round

- The HR interviewer introduced herself and then asked me to introduce myself.
- She asked me about my experience in the first and second rounds.
- She asked what my favorite project is and why.
- She asked me to explain what I know about Tekion.
- I explained the projects I had built, the difficulties I faced during those projects, and how I overcame them.
- She asked why I chose Tekion.
- At the end, I asked her the same questions I had asked in Round 2. With her permission, I also asked a third question (which I don't fully remember).

Takeaways:

- Strong knowledge in **DSA, OS, CN, DBMS, and OOPS** is required.
- Prepare your **self-introduction** well. Try to include your interests, the tech stacks you know, the projects you have done, and any achievements if you have them.
- For DSA questions, don't jump directly to the optimal solution. First, explain the **brute force solution** along with its time and space complexity, then move to the **optimal solution**, highlighting why it is better and how you arrived at the intuition. Even if you don't know the exact solution, share your thought process and intuition.
- Be well-prepared on your **projects**—know them in detail, including challenges faced and how you solved them.
- Always prepare a few **questions to ask the interviewer** at the end of the round.
- Be **confident** and Be prepared

Results were announced on the same day

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